

Detecting Plant Disease with Machine Learning

Abstract

Global food insecurity is significantly exacerbated by crop diseases, which currently account for an estimated 20–40% loss in annual yield. While existing deep learning models using leaf images have shown high accuracy, they remain inherently reactive, identifying symptoms only after physiological damage is visible. This research proposes a more comprehensive multimodal machine learning system that fuses a custom Convolutional Neural Network (CNN) for image analysis with numerical soil data, such as pH, moisture, and NPK levels. By analyzing a dataset of over 17,600 images across 13 distinct plant classes (including Tomato, Potato, and Pepper), this study tests the hypothesis that a fused model will achieve a higher F1 score than single-modality models by capturing latent environmental stressors that precede disease.

Objectives

The primary objective of this study is to develop a dual-branch neural network architecture capable of processing spatial image data and numerical soil features simultaneously to create a holistic diagnostic tool. By integrating specific soil parameters, Nitrogen, Phosphorus, Potassium, pH, moisture, and electrical conductivity, the research evaluates whether environmental context can significantly improve diagnostic accuracy over traditional vision-only methods. Additionally, an ablation study is conducted to compare the performance of the "Image-Only" versus "Multimodal" systems, aiming to quantify the added value of environmental data in increasing the precision and recall of agricultural AI.

Discussion

The results of this study support the hypothesis that multimodal systems provide a superior framework for plant pathology compared to single-modality CNNs. By accounting for the multidimensional drivers of plant health, this research bridges the gap between image classification and proactive agricultural management. Future work will focus on deploying this system as a mobile application using SwiftUI and CoreML to provide farmers with real-time field diagnostics. Additionally, there are plans to scale the model by integrating UAV-based remote sensing and real-time weather API data, transitioning the technology from a diagnostic tool into a predictive system capable of identifying disease risks before yield losses occur.

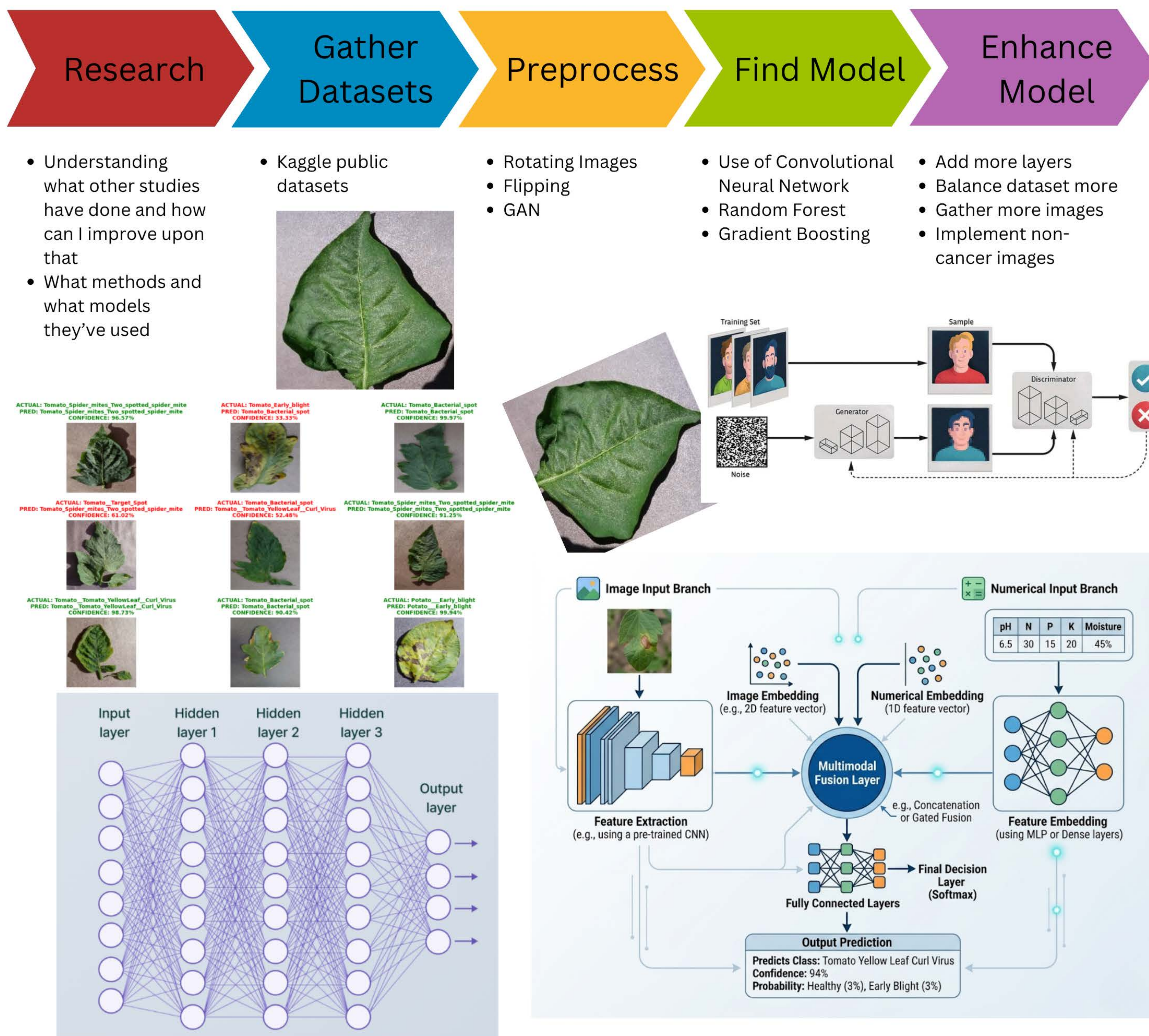
Plant_ID	Soil_Moisture	Ambient_Temperatu	Soil_Temperature	Humidity	Light_Intensity	Soil_pH	Nitrogen_Level	Phosphorus_Level	Potassium_Level	Chlorophyll_Content	Electrochemical_Si	Plant_Health_Status
1	27.52110877	22.24024536	21.90043536	55.2919039	556.1728051	5.581954516	10.00364972	45.80685203	39.07619903	35.70300571	0.9414021465	High Stress
1	14.83556615	21.70676328	18.68089194	63.94918051	596.1367212	7.135704906	30.712562	25.3943926	17.94482607	27.99329577	0.1648993611	High Stress
1	17.08636197	21.18094556	15.39293913	67.83795649	591.1246268	5.656852249	29.33700224	27.57389185	35.70653021	43.64630752	1.081728041	High Stress
1	15.33615608	22.50330104	22.77830339	58.10081101	241.4124764	5.584522687	18.06662065	28.18070481	28.25774575	37.83809483	1.186087071	High Stress

Method

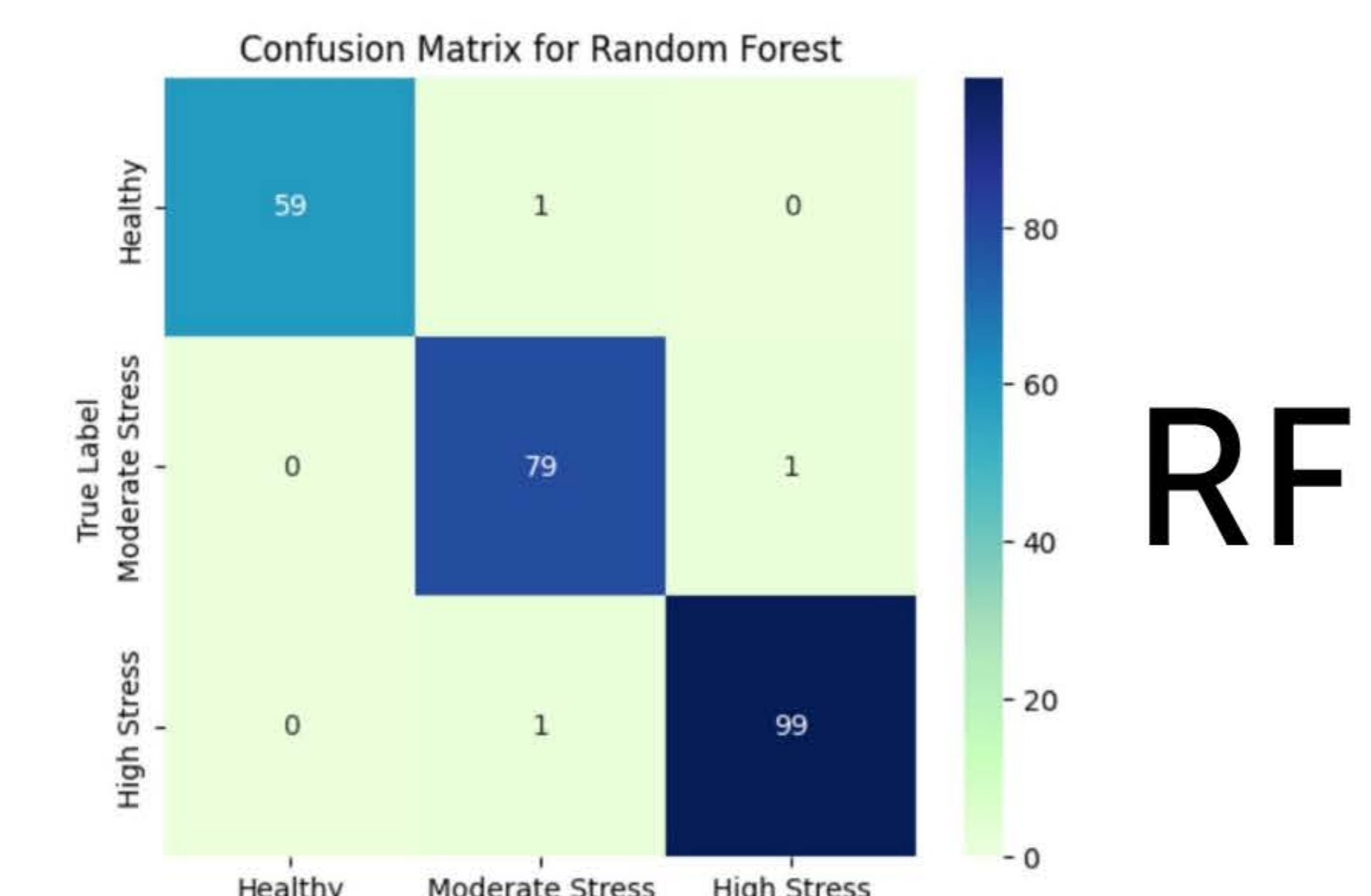
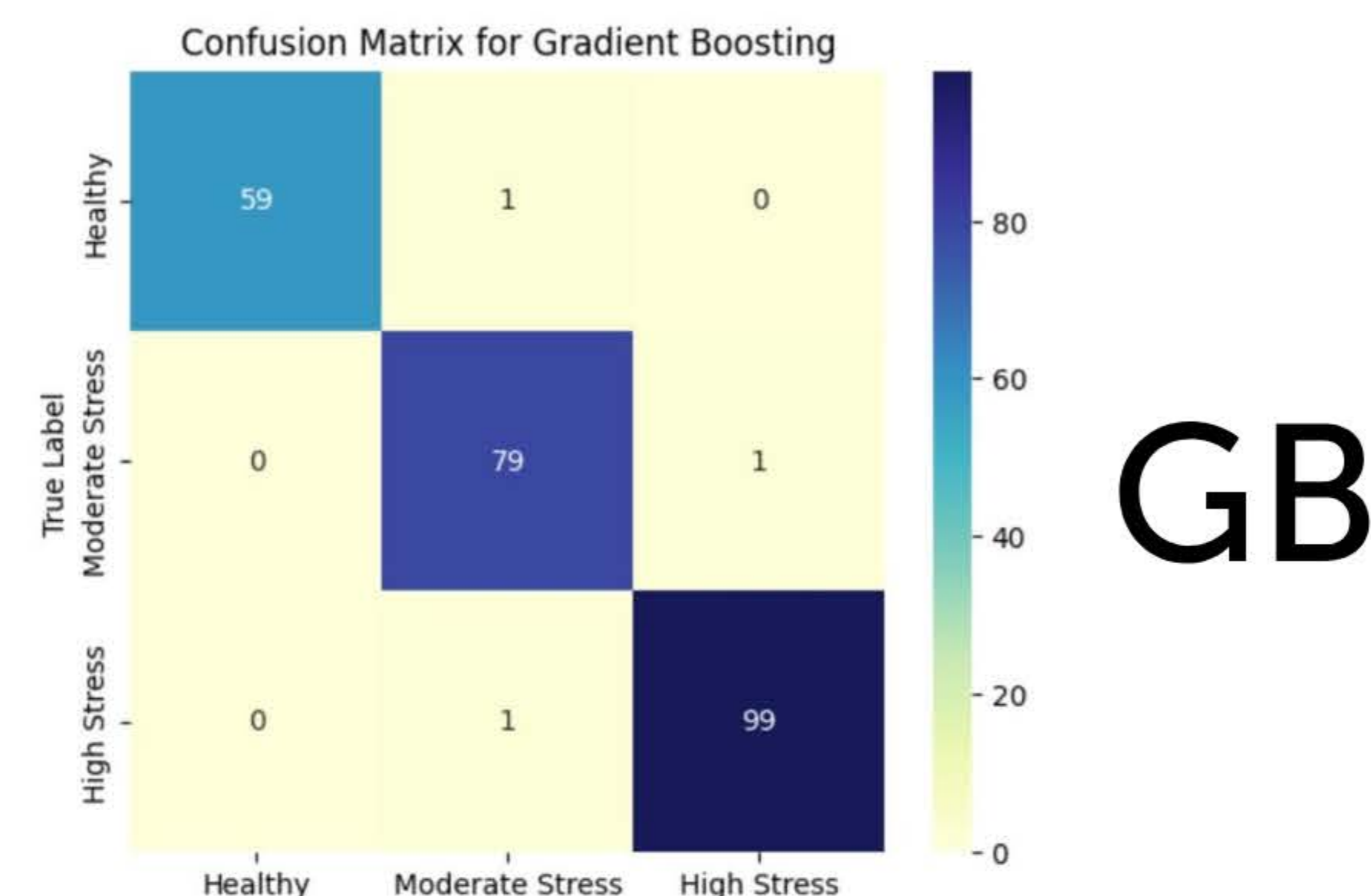
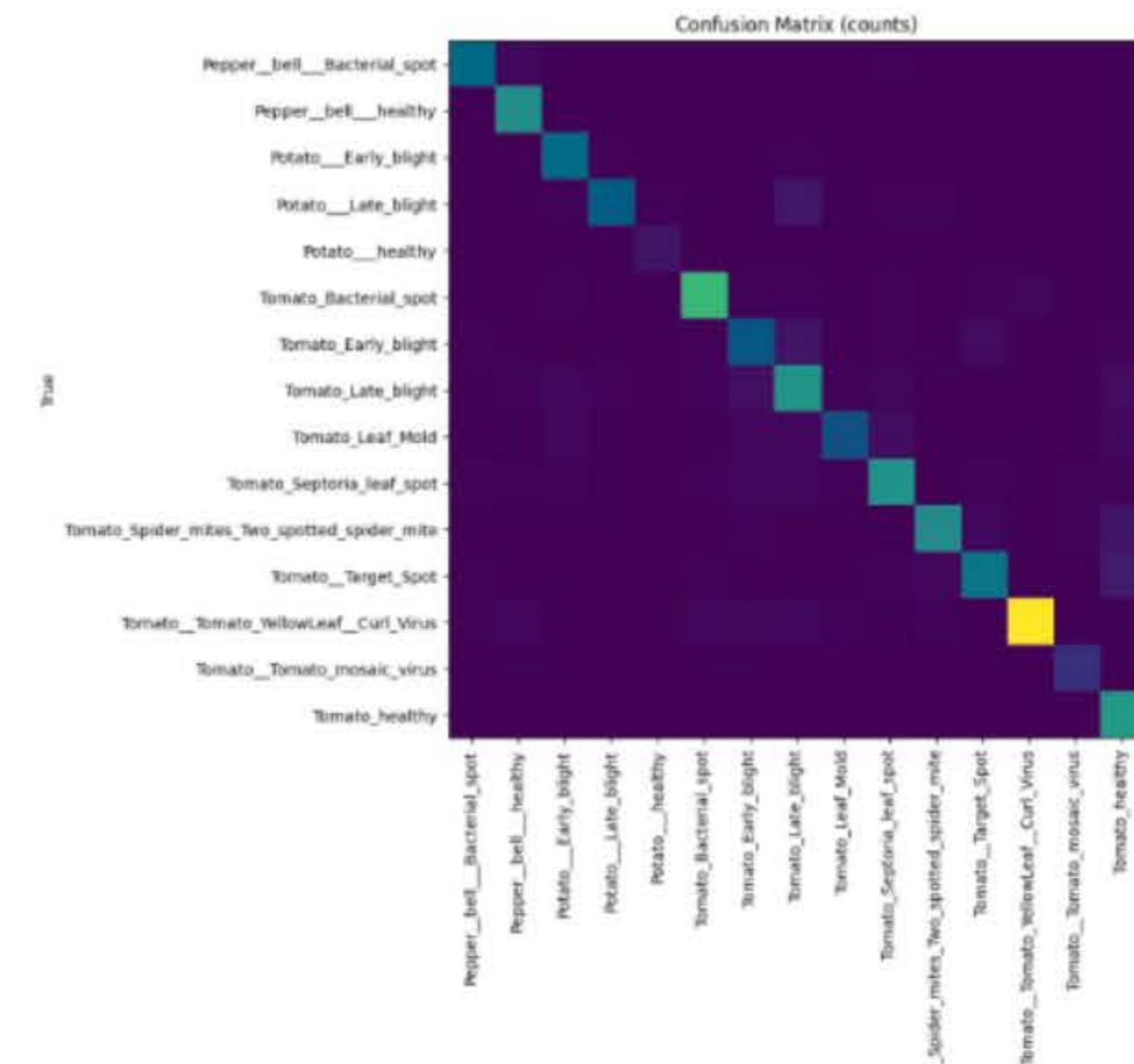
This research project began by addressing the limitations of single-mode plant pathology models, which often lack the environmental context necessary for field accuracy. To improve generalizability, this study integrated leaf imagery with numerical soil parameters (pH, moisture, NPK), creating a more comprehensive dataset than traditional vision-only approaches.

The technical implementation utilized a custom TensorFlow/Keras pipeline. Preprocessing included an automated validation script to remove corrupt files and standardizing images to 128x128 pixels. To reduce model bias, data augmentation and a Generative Adversarial Network (GAN) were used to balance the dataset across 13 distinct classes, resulting in over 17,600 total samples.

The model architecture features a dual-branch neural network. The vision branch uses a Sequential CNN to extract spatial features, while the numerical branch processes soil data via a Multilayer Perceptron (MLP). These branches are fused using a late-fusion layer and optimized with Early Stopping to prevent overfitting. Performance was rigorously evaluated using weighted F1 scores and confusion matrices to ensure reliable classification.



Results



Model Configuration	Input Modality	Accuracy	Weighted F1-Score
Sequential CNN	Image Data Only	0.892	0.886
Random Forest	Soil Data Only	0.981	0.98
Gradient Boosting	Soil Data Only	0.994	0.993
Multimodal Fusion (CNN + MLP)	Images + Soil Data	0.997	0.996

References

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